MAHDI BEDJA

AI Engineer
Paris, France · +33 7 45 60 96 80 · mahdibedja@gmail.com
linkedin.com/in/mahdi-bedja-0b94031b3 · github.com/MehdiBedja

PROFILE

AI engineer graduated from ESI (Algiers), currently pursuing a Master 2 in Intelligent Systems at **Sorbonne Paris Nord University**. I have **10 months of applied AI** experience in medical software (imaging, speech recognition, embeddings) and an end-to-end mobile AI project, from model training to Android deployment. Seeking a **final-year internship (PFE)** in applied AI, available from March 1, 2027.

PROFESSIONAL EXPERIENCE

AI Engineer – BM TECH
Nov 2025 – Aug 2026 | Algiers, Algeria

Leading Algerian company providing software for medical laboratories and public and private hospitals

- **Medical imaging:** pipelines for detection, segmentation and classification of 14 lung pathologies on chest X-rays (YOLO, CNN); independent bone fracture detection pipeline.
- **Medical ASR in French:** built a synthetic speech dataset with Qwen3 (voice cloning of physicians from real patient radiology reports), then fine-tuned Whisper: **3% WER** on a held-out synthetic medical test set.
- **Real-time medical dictation:** Dragon Medical-inspired system built on WhisperLiveKit / faster-whisper (streaming), enhanced with fuzzy matching and voice commands: **8% WER** in live streaming.
- **Agentic RAG & patient document Q&A:** architected an Agentic RAG system using LangChain and locally hosted Qwen models, querying over patient records and clinical documents.
- **Medical interoperability (R&D):** dense embeddings of the LOINC dataset with Qwen for semantic matching of lab test results across hospitals.
- **Industrialization:** containerized services with Docker, structured logging and built-in monitoring.

Engineering Intern – CM Consulting
Jul 2024 – Aug 2024 | Algiers, Algeria

- Python tool for extracting and structuring data from PDF tenders (spaCy, pdfplumber), with a Tkinter interface and automatic generation of formatted reports.

EDUCATION

Master 2 – *Intelligent Systems*
Sep 2026 – Present | Paris, France

Sorbonne Paris Nord University

Engineer's Degree & Master's in Computer Science
Sep 2020 – Sep 2025 | Algiers, Algeria

ESI – National Higher School of Computer Science

PROJECTS

Mobile Agricultural Pest Detection System (Thesis Project)
Sep 2024 – Sep 2025

- Conducted applied AI research benchmarking object detection architectures (YOLOv7–v11, SSD, Faster R-CNN) on custom datasets with advanced data augmentation strategies to optimize accuracy-latency tradeoffs.
- Quantized and exported the optimal YOLOv11s model (93.8% mAP@50) to TFLite for edge deployment, embedding it within an Android app (Kotlin, Jetpack Compose) enabling real-time offline inference.
- Engineered a scalable Django, PostgreSQL, and Cloudinary backend deployed on Render for data persistence and user management.

Intelligent Legal Monitoring System with LLM Chatbot
Feb 2024 – Jun 2024

- LLM-powered chatbot answering questions on legal texts, on a Django + React platform with Selenium scraping of new laws, Elasticsearch advanced search and push notifications to subscribers.

Automated PostgreSQL Deployment and Testing with Jenkins
Mar 2024 – Jun 2024

- Deployment on Render, Flyway migrations, pgTAP tests and an automated Jenkins CI/CD pipeline.

TECHNICAL SKILLS

Generative AI & LLMs: Agentic RAG, LangChain, LLM Agents, Local LLM Inference (Qwen), Dense Embeddings

Speech & Vision: Whisper (Fine-tuning & Streaming), ASR/TTS, Object Detection & Segmentation (YOLO, SSD, Faster R-CNN)

ML Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face, TFLite, spaCy

Backend & Mobile: Python, Django, REST APIs, PostgreSQL, Elasticsearch, Kotlin (Jetpack Compose)

MLOps & Infrastructure: Docker, CI/CD (Jenkins), Git, Structured Logging, Monitoring, Render

LANGUAGES & OTHER

Languages: English C1 (IELTS 7.5) · French C1 (TCF C1) · Arabic (native)

Interests: Applied AI & GenAI Research, Algorithmic Problem Solving, Open Source